

Trainings ▾

General Information

Fuel Tubes

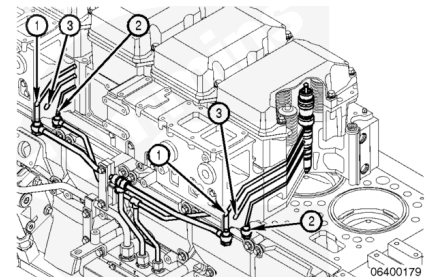
The QSK23 fuel rail supply (1), timing rail supply (2), and fuel drain (3) are integrated into the intake manifold housing.

Fuel supply and drain lines are sized to provide sufficient flow for injection and timing functions. Straight thread face o-ring fittings are used for superior leak prevention.

The recommended fuel pump inlet line size provides a maximum clean filter restriction of 102 mm Hg [4 in Hg] at high idle-no load.

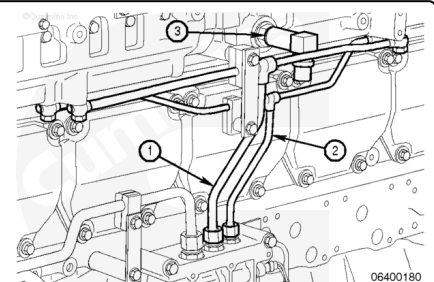
The fuel drain fitting provides a maximum drain line restriction of 203 mm Hg [8 in Hg].

The fuel temperature at the fuel pump inlet **must not** exceed 71°C [160°F]. A fuel cooler in the fuel drain circuit will possibly be required. The fuel cooler is to be provided by the original equipment manufacturer (OEM).



The fuel rails in the intake manifold receive fuel from the Electronic Control Valve Actuator (ECVA) through a fuel rail supply tube (1) and fuel timing supply tube (2). The fuel then passes through the fuel rails, through the cylinder head, and into the injectors. Unused fuel flows back through the cylinder head, into the fuel rail, and out through the fuel drain fitting (3) to return to the fuel tank.

Industrial applications with dual intake manifolds will have a steel return line connecting the two manifolds to the fuel drain fitting.



Quantum™ Injector

The QSK23 Quantum™ injector begins operation with metering. The lower plunger retracts during the inner base circle portion of the camshaft profile, thus uncovering the rail feed port. The fuel is Pressure Time (PT) metered into the nozzle. The

amount metered is a function of rail pressure and engine speed. This supply pressure will be as high as 1379 kPa [200 psi] at maximum fueling/speed and as low as 13.8 kPa [2 psi] at idle.

The timing fuel is also PT metered through a separate metering port into a chamber between the upper and timing plungers in the barrel. This also occurs during the inner base circle portion of the camshaft profile beginning when the upper plunger retracts and uncovers the timing feed port. Timing metering ends when the camshaft drives the upper plunger downward, covering the metering feed port and trapping the fuel between the upper and timing plungers. This trapped fuel acts as a solid hydraulic link. The lower plunger also moves downward, closing the rail feed port and ending injected fuel metering.

The amount of fuel metered into the timing chamber determines the amount of separation of the upper and timing plungers. This separation amount, called overtravel, varies from an approximate minimum of 2 mm [0.078 in] at torque peak fueling to a maximum of 9 mm [0.354 in] at high idle fueling. The amount of overtravel is changed to vary the start of injection for all speed and fueling conditions.

